MANTEY JOSHUA KOJO ADUAMPONG

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EDUCATION

Northeastern University, Boston, MA | Advisor: Dr. Srirupa Chakraborty. | Sept 2024 – Present
Doctor of Philosophy in Chemical Engineering

Northeastern University, Boston, MA | Sept 2024 – May 2026
Master of Science in Biosystems and Biomaterials

Kwame Nkrumah University of Science and Technology, Ghana | Sept 2019 – Sept 2023
Bachelor of Science in Petrochemical Engineering Class Rank: **5th** /87

TECHNICAL SKILLS

Programming & Data: Python (Scikit-learn, Pandas, NumPy, Matplotlib), MATLAB, MongoDB

Machine Learning & Statistics: MongoDB, Supervised Learning, Model Validation, Cross-Validation, Regression Analysis, Hypothesis Testing, Sensitivity Analysis

Computational Tools: Molecular Dynamics (AMBER, NAMD, Schrödinger, VMD, PyMOL, Chimera), Process Simulation (Aspen HYSYS, ANSYS Fluent), HPC Computing (NSF ACCESS)

PROJECT EXPERIENCE

Graduate Research Assistant Computational Biology & AI Applications | Jan 2025 – Present
SimBioSys Laboratory, Northeastern University

- Conducted large scale MD simulations of **3+ million** atom MUC2 mucin bile salt systems generating **>10 TB** trajectory data modeling molecular interactions governing interactions through glycosylated mucosal barriers in oral delivery applications.
- Analyzed simulation trajectories using Python (MDAnalysis, NumPy, SciPy) to quantify mucin bile salt interactions enabling data driven formulation design.

Air Quality Forecasting System | Nairobi Environmental Monitoring | Jan 2026 - Feb 2026
Python • MongoDB • Machine Learning • Time Series Analysis

- Engineered end to end ML pipeline processing **200,000+** hourly air quality measurements across Nairobi stored in MongoDB, developing ensemble time series achieving **RMSE < 15 µg/m³** for 24hour PM2.5 forecasts and **82%** accuracy predicting hazardous air quality events.
- Built automated ETL workflow using MongoDB aggregation framework to extract temporal patterns and engineer **50+** features, integrating external datasets and reducing prediction error by **40%** compared to baseline models, enabling data driven public health early warning system for vulnerable population.

Housing Price Prediction System | Mexico City Real Estate Analytics | Nov 2025 - Dec 2025

- Developed predictive ML model processing **10,000+** real estate listings across Mexico City's Distrito Federal, implementing Ridge regression to forecast apartment prices for data driven real estate investment and market analysis applications.
- Engineered robust preprocessing pipeline handling missing data **15%** of records, extracting hierarchical location features from unstructured address strings, to reduce feature space noise by **40%**.

Research Collaborator – HIV Vaccine Development | May 2025 – July 2025
Fred Hutchinson Cancer Center

- Analyzed **8+** conformational states from MD simulations and cryoem structures of HIV envelope proteins using Python to identify conserved binding pockets for broadly neutralizing antibody development.

Computational Modeling Projects | Oct 2024 – May 2025
Northeastern University

- Built ML models analyzing **50,000+** atomic configurations from MD simulations to predict CO_2 adsorption on NH_4HCO_3, informing material optimization for industrial scale carbon capture.
- Processed CFD simulation data (1M+ mesh elements) modeling natural gas flow, heat transfer in underground storage caverns, optimizing injection parameters to enhance storage efficiency and reduce fracture-induced leakage risks.

Bioprocess Engineering : Data Driven Optimization | Sept 2024 – Dec 2024
Northeastern University

- Achieved 250 mg/L recombinant protein yield by analyzing time series bioreactor data across 5+ fermentation runs, applying statistical process control and DOE methods aligned with FDA 21 CFR Part 211
- Implemented PAT framework with real time sensor monitoring to optimize IPTG induction reducing batch variability by 30% through data driven process adjustments.

Science & Engineering Fair (2025) | Member, American Institute of Chemical Engineers

WORK EXPERIENCE

Data Driven Chemical Engineer (Part Time) | 08/2021–12/2024
Lee House and Chemical Ventures | Accra, Ghana

- Engineered data analytics pipeline processing 10,000+ production records (batch parameters, quality metrics, equipment performance) from food manufacturing operations serving local and export markets, applying machine learning techniques (regression analysis, anomaly detection) to optimize process efficiency and predict quality outcomes
- Developed predictive quality model analyzing historical batch data (ingredient compositions, processing conditions, environmental factors) achieving 85% accuracy in forecasting product quality deviations before completion, enabling proactive adjustments that reduced waste by 18% and ensured export quality compliance
- Built demand forecasting system using time-series analysis and statistical modeling to process sales data across 20+ SKUs and regional markets, optimizing production planning to reduce inventory costs by 15% while maintaining 98% order fulfillment rates for international clients
- Applied data mining and root cause analysis techniques to equipment sensor data (temperature, pressure, flow rates) identifying patterns associated with downtime events, implementing predictive maintenance strategies that reduced unplanned stoppages by 25% and improved overall equipment effectiveness (OEE)

Data-Driven Quality Control & Analytics Intern | Oct 2021 – Dec 2021
Ghana Oil Company | Accra, Ghana

- Conducted nationwide fuel quality audits across 50+ retail stations, analyzing 500+ fuel samples using ERASPEC fuel analyzers and gas chromatography (GC-FID) to ensure compliance with Ghana Standards Authority specifications and environmental emission standards, identifying 15+ non-conformance cases requiring corrective action.
- Developed data-driven quality monitoring framework processing test results (octane ratings, sulfur content, aromatic compounds, vapor pressure) from multiple testing sites, applying statistical analysis to identify regional quality trends and detect anomalies that could indicate supply chain issues or adulteration.
- Engineered fuel blending optimization model analyzing compositional data from 100+ blending operations, using regression analysis to predict final fuel properties (RON/MON ratings, density, volatility) and identify cost-effective formulations that reduced production costs by 8% while maintaining performance specifications.
- Built automated quality reporting system consolidating GC analysis data and field test results into actionable insights, generating statistical process control charts and identifying systematic deviations that enabled proactive quality interventions across the distribution network.

ORGANIZATIONS

- Member, International Association of Engineers (IAENG), Boston, MA
- Member, American Institute of Chemical Engineers, USA.
- Member, Biophysics Society of USA
- Member, African Graduate Students Association, Northeastern University USA
- Member, Ghana Institution of Engineering, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
- Member, Ghana Engineering Student Association
- Assistant Audit Committee Head, Petrochemical Engineering Students Association

JUDGING EXPERIENCE

- Judge, Massachusetts Science and Engineering Fair, Clark University. 05/2025
- Judge, Massachusetts Science and Engineering Fair, Worcester Polytechnic Institute. 04/2025.

- Judge, Sigma Xi Symposium, Undergraduate in Northeastern University. Faculty of Chemical Engineering. 04/2025.

GRANTS

- **NSF ACCESS Award** for Computational Research, August 2025
- **Grant Recipient**, New England African Business Expo, Worcester, Massachusetts. May 2025

AWARDS

- **Most Innovative intern,** Lee House & Chemical Ventures, 2022/2023
- **Best Hardworking Employee**, Lee House & Chemical Ventures, 2021/2022
- **Excellent Student Award**, Provost of College of Engineering, KNUST, 2020/2021
- **Excellent Student Award**, Provost of College of Engineering, KNUST, 2019/2020

PRESENTATIONS AND CONFERENCES

- **Mantey Joshua Kojo.A** , Simran Pandey, Rebecca Crrier, Srirupa Chakraborty, "Enhanced Sampling to Study Interactions with Bile Salts", AICHE Annual Meeting-Boston. November 2025
- Atombeat Conference and Learn Series. July 2025
- Festival of Genomics and Biodata. June 2025
- **Mantey Joshua Kojo.A.**, Pandey, S., Chakraborty, S.," Mucosal Interactions with Signaling Molecules", Professional Development, Poster Presentation, Department of Chemical Engineering, Northeastern University. April 2025
- **Mantey Joshua Kojo.A**, "Pyrolysis and Carbonization as a Fuel Preparation Method", Undergraduate Poster presentation, Department of Chemical Engineering, KNUST. March 2025

PUBLICATIONS

- Amevor, R., **Mantey Joshua Kojo Aduampong**, Nketiah, T.A, "Evaluation of Ai-Based Predictive Models for Early Cancer Detection: Statistical Considerations and Validation Methods." Jr.Inn.Sci 2.1(2026):pp 1-10.
- **Mantey Joshua Kojo.A,** Muntaka Is-mail, "Economic Evaluations of Advanced Drug-Delivery Platforms:Cost-Effectiveness Evidence in U.S Disease Contexts".-SARC Publisher, 2025.
- **Mantey Joshua Kojo. A**, Benjamin Edem Meteku, Eric Baffoe, "Amino Acids as Corrosion Inhibitors"- Taylor & Fracis group, 2025.
- Timothy Kizza, **Mantey Joshua Kojo. A** , Felix Kaiser, "Survival Analysis in U.S chronic disease research: A systematic review of methods and applications". IJFLRS 2025
- Agyei, D., Owusu-Kwarteng, J., Kehinde, B., Ohemeng-Boahen, G., **Mantey Joshua Kojo.A**. "The Role of Indigenous Foods in Achieving Food Security in Africa", 2024.

MEDIA PUBLICATIONS

https://www.myjoyonline.com/joshua-mantey-the-use-of-ai-in-drug-delivery/

https://www.myjoyonline.com/digital-intelligence-environmental-disaster-ais-hidden-climate-crisis/